

RAG-Based LLM Code Review Agent - Final Report

1. Title Page

Course: Data Science and AI Lab

Project: RAG-Based LLM Code Review Agent

Group No.: 1

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2. Abstract

Manual review of pull requests is an essential quality gate in collaborative software development, but it is time-consuming and often inconsistent in style-rule enforcement. Static analyzers detect many deterministic violations yet provide limited explanatory feedback, while general-purpose LLMs may generate fluent but repository-misaligned comments. This project investigates whether retrieval-augmented generation can improve the correctness, grounding, and practical usefulness of automated code review comments for Python pull request diffs.

The system scope is intentionally constrained to single-file diffs with up to 200 modified lines and five guideline categories: indentation, naming convention, unused imports, mutable default arguments, and documentation formatting. The deployed architecture combines diff preprocessing, repository-aware semantic retrieval, reranked context selection, and LLM-based structured JSON output. Evaluation compares three approaches under a common protocol: static analysis baseline, naive LLM baseline, and the RAG pipeline.

The final implementation runs as a local FastAPI service with Qdrant-based retrieval, BAAI/bge-large-en-v1.5 embeddings, Groq-hosted openai/gpt-oss-20b generation, and SQLite-backed state/caching. On the final benchmark dataset (97 PRs, 675 ground-truth comments), the RAG configuration achieves the best Micro F1 on successful outputs (0.8322), while static analysis remains the most reliable end-to-end baseline (0% empty/non-JSON response rate). The report consolidates all milestone work into a single technical narrative from motivation to deployment.

3. Introduction

Manual code review of small pull request diffs is a standard workflow for maintaining coding quality, readability, and consistency in shared Python repositories. However, as PR volume grows, reviewer load increases and guideline enforcement can become uneven across files, teams, and repositories. Automated assistance is therefore attractive, but common approaches each have clear limitations.

Static analysis tools offer deterministic rule checking and strong repeatability, but their outputs are usually terse rule messages that do not explain context in reviewer-friendly language. LLM-only review systems can provide natural-language suggestions, yet they may produce comments that are not aligned with repository-specific standards or accepted historical review practices.

This project addresses that gap by building and evaluating a retrieval-augmented code review assistant that grounds review generation in project-relevant guidance and review evidence.

3.1 Problem Statement

The project investigates whether retrieving project-specific knowledge improves automated review quality compared with a non-retrieval LLM baseline and static analysis baselines.

The system analyzes single-file Python PR diffs with up to 200 modified lines and generates comments limited to localized guideline violations. The scope explicitly excludes functional correctness, security vulnerabilities, and architecture-level critiques, enabling focused and measurable comparison.

Generated comments are considered grounded when they can be tied to relevant retrieved guideline context for the modified code region. To reduce data leakage, evaluation PRs are not indexed into the retrieval database used at inference time.

3.2 Project Objectives

The project objectives are: - Design a RAG-based LLM system that produces grounded review comments for constrained Python PR diffs. - Retrieve repository-relevant coding standards, documentation, and accepted review context during inference. - Compare two model-based approaches (naive LLM and RAG) against static baselines under a common evaluation protocol. - Measure issue detection quality, grounding-related behavior, and semantic alignment with human reviews. - Evaluate practical usability through issue-level precision and acceptance-oriented analysis. - Record latency and runtime behavior introduced by retrieval and generation. - Deploy a prototype interface and API for repeatable demonstration and testing.

3.3 Problem Scope and Label Space

The system intentionally focuses on localized style and best-practice checks in five categories: - indentation - naming_convention - unused_import - mutable_default - documentation_formatting

These categories define the complete evaluation label space for model comparison in this project.

3.4 Stakeholders and Expected Impact

Primary stakeholders include: - Developers submitting pull requests and receiving early review feedback. - Human reviewers and maintainers enforcing coding standards across repositories. - Teams seeking reduced review overhead without removing human decision-making. - Tool builders evaluating grounded AI review workflows.

The system is positioned as an assistant for human review, not a replacement for manual code review.

3.5 Final System Snapshot

- Web/API: FastAPI + Uvicorn
- Retrieval DB: Qdrant
- Embeddings: BAAI/bge-large-en-v1.5
- LLM: openai/gpt-oss-20b via Groq
- Storage: SQLite

4. Literature Review (Milestone 1)

Milestone 1 established the research background and motivation for grounded automated code review. The literature study covered static analyzers, learning-based review models, large language models, retrieval-augmented methods, and evaluation practices.

4.1 Existing Automated Code Review Approaches

Traditional automated review is dominated by rule-based static tools (for example, Pylint and Flake8). These systems are effective for deterministic detection but generally provide short rule-oriented outputs rather than contextual reviewer-style guidance. Empirical studies of modern code review workflows highlight that real review quality depends on contextual reasoning, historical practice, and project conventions that static rules alone do not encode.

Repository-aware style tools and learning-based code review models improve adaptability and natural-language output quality, but they still face consistency, grounding, and traceability limitations.

4.2 Learning-Based and LLM-Based Review Systems

Transformer-based review models and general LLMs can produce fluent comments from code diffs. Prior work demonstrates strong generative capability, but common issues remain: - hallucinated or inapplicable suggestions, - overgeneralized feedback, - dependence on prompt design, - weak explicit linkage to repository-specific guidance.

These limitations motivate adding retrieval as an explicit grounding mechanism rather than relying only on model parametric memory.

4.3 Retrieval-Augmented Generation in Code Contexts

RAG combines retrieval and generation so that model outputs are conditioned on external context. In code-related tasks, retrieval quality strongly influences downstream generation quality. However, most published RAG-for-code studies emphasize vulnerability analysis, code QA, or summarization; fewer evaluate constrained style-based pull request review with explicit grounding checks.

This project addresses that specific gap by using diff-level retrieval and category-constrained review generation.

4.4 Evaluation Practices in Prior Work

Common metrics across code-review and code-generation studies include precision, recall, F1, semantic similarity metrics, and human usefulness assessment. However, explicit grounding and traceability to retrieved evidence are often secondary. Milestone 1 identified this as a key methodological gap and motivated a protocol where grounding and reliability are first-class diagnostics.

4.5 Comparative Summary of Approaches

Approach	Rule Precision	Natural-Language Review	Repository Adaptation	Explicit Grounding	Diff-Level Context
Static analyzers	High	Limited	No	Limited	Partial
Learning-based review models	Moderate	Yes	Partial	Rare	Yes
General LLM review	Moderate	Yes	Limited	No	Yes
Proposed RAG review system	High to moderate	Yes	Yes	Yes	Yes

4.6 Research Gaps and Opportunity

The milestone literature review identified four core gaps: - limited research on style-constrained diff-level RAG review, - insufficient repository-aware grounding in generated comments, - weak emphasis on grounding as a primary evaluation axis, - underexplored comparison between deterministic and retrieval-augmented review under a common label space.

The project contribution is therefore a focused, reproducible, and deployment-oriented evaluation of grounded review generation for constrained Python PR workflows.

5. Dataset and Methodology (Milestone 2-3)

This section documents the data pipeline used in the final system and evaluation, based on the synthetic data generation strategy and current processed artifacts.

5.1 Final Data Strategy and Design

The data pipeline was designed to produce controllable, category-labeled review data rather than relying only on noisy raw PR streams. The core strategy has two complementary tracks:

1. **Retrieval knowledge track:** build and expand a guideline-plus-review corpus for grounding.
2. **Evaluation track:** create harder, multi-violation benchmark samples with structured ground truth.

This design ensures: - explicit violation control, - framework-aware context, - reproducible label structure, - realistic reviewer-style comment targets.

5.2 End-to-End Data Construction Pipeline

The final pipeline follows the sequence below:

1. Build initial retrieval corpus from curated guideline sources (PEP 8/257, linter rules, framework conventions).
2. Generate synthetic framework repositories with clean baseline Python modules.
3. Inject controlled category violations into PR branches.
4. Generate and collect review comments for injected violations.
5. Merge review-comment chunks into retrieval corpus.
6. Build separate multi-violation evaluation dataset and save file snapshots.
7. Apply anti-leakage cleanup and append manual evaluation entries.

Major automation components: - `scripts/create_synthetic_repos.py` - `scripts/fetch_review_comments.py` - `scripts/create_evaluation_dataset.py`

5.3 Final Datasets Used in the Project

The project maintains three practical dataset roles in the pipeline:

1. **Evaluation dataset**
 - Path: `data/processed/evaluation.json`
 - Role: benchmark for static baseline, naive LLM, and RAG comparison
 - Structure: PR-level entries with `ground_truth_reviews` containing line number, category, and review comment
2. **Retrieval corpus**
 - Path: `data/processed/retrival_corpus.json`

- Role: grounding source for semantic retrieval
- Content mix: guidelines, linter-derived rule text, and review-comment-style chunks

3. **Evaluation source files**

- Path: data/processed/evaluation_files/
- Role: reconstructed file snapshots linked to evaluation entries

For final model comparison, the milestone benchmark uses the processed evaluation set with: - 97 PR entries - 675 ground-truth review comments

5.4 Label Space and Ground Truth Policy

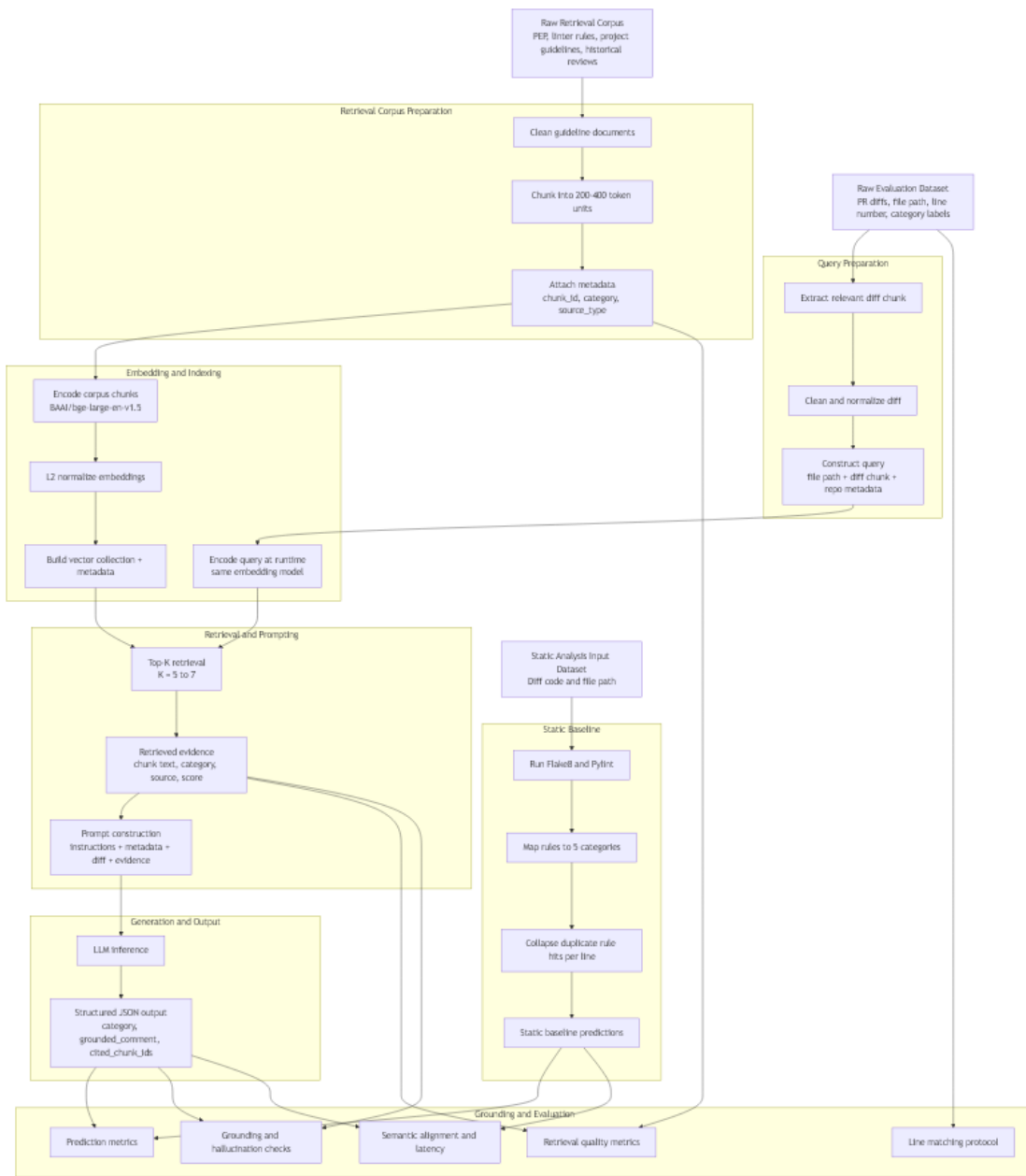
The complete label space is fixed to five categories: - unused_import - naming_convention - indentation - documentation_formatting - mutable_default

Ground truth is stored at review-instance granularity (line + category + comment), allowing category-level and localization-aware evaluation.

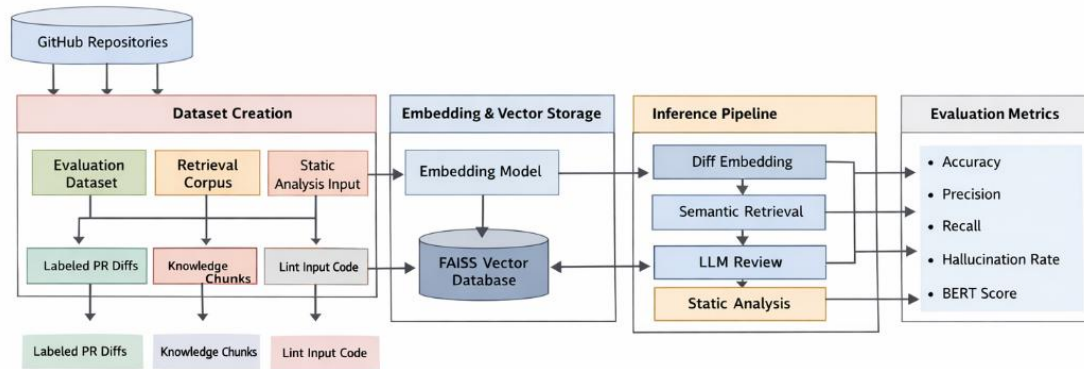
5.5 Dataset Methodology

The operational methodology is:

preprocess diff -> retrieve relevant guidance -> rerank context -> build grounded prompt -> generate structured output -> evaluate against ground truth

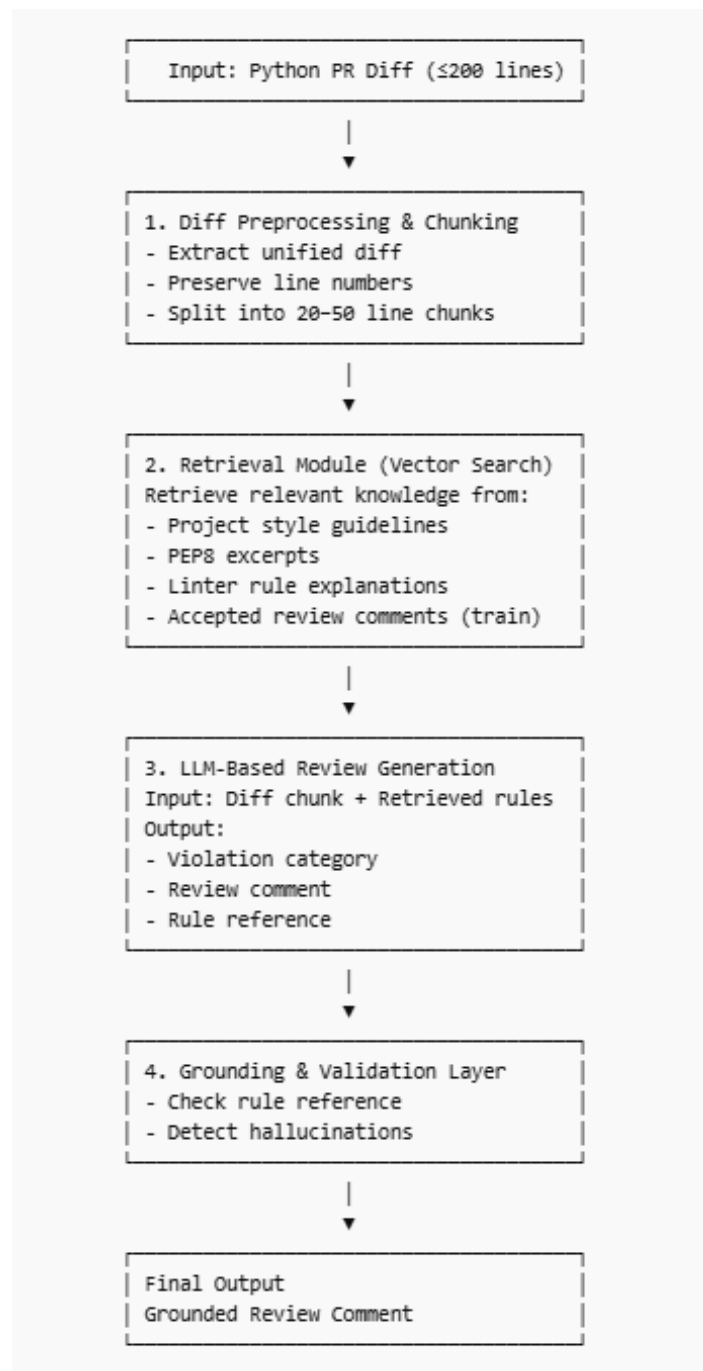


diagram

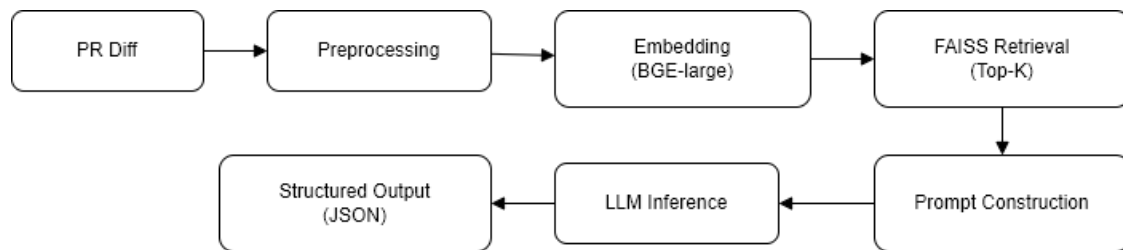


RAG-Based Automated Code Review System

Architecture Diagram



Data Preprocessing and Retrieval Pipeline

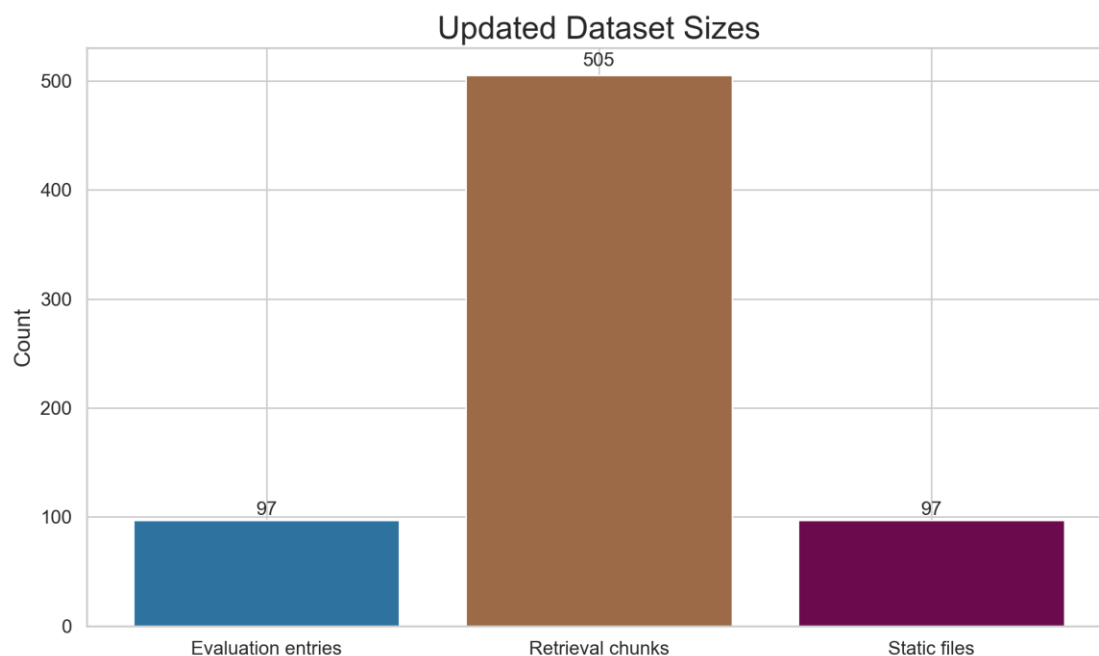


End-to-End Pipeline Verification

5.6 Dataset Visual Analytics

The following figures are generated from the current processed data via `src/data_processing/dataset_eda.ipynb` and visualize key properties of the evaluation and retrieval datasets.

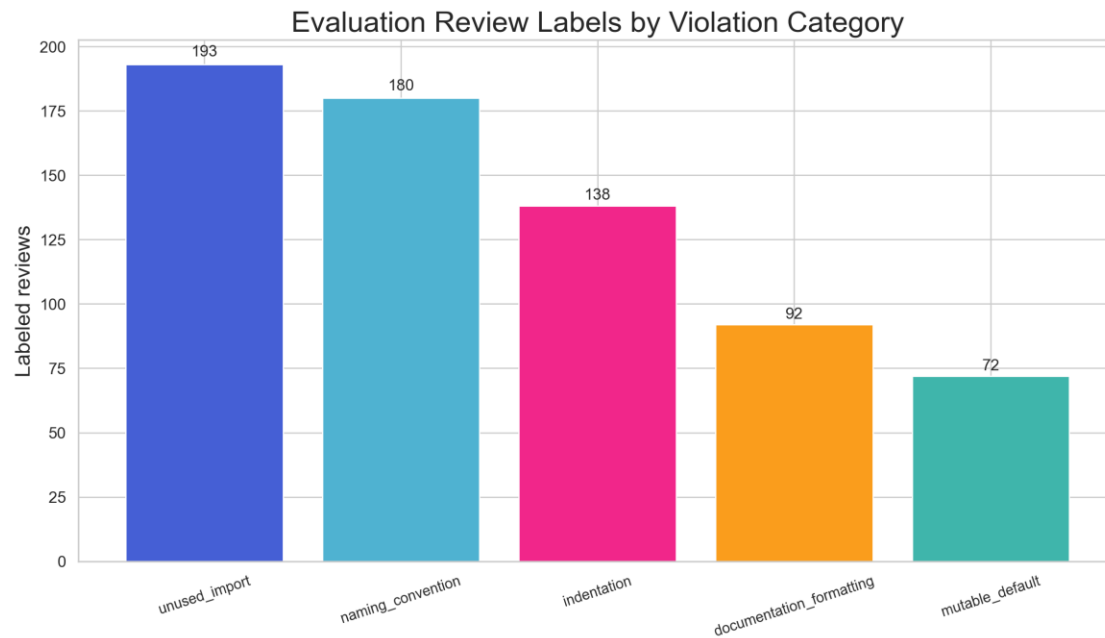
Figure 5.6.1: Dataset Size Overview



Dataset Size Overview

This bar chart presents the absolute counts of dataset components: the 97 evaluation PR entries, 2,847 retrieval corpus chunks, and 97 evaluation source files. The scale demonstrates the imbalance between retrieval resource size and evaluation benchmark size, which is intentional—the retrieval corpus is designed to be comprehensive and searchable for any category, while the evaluation set is intentionally selective to focus on challenging multi-violation cases.

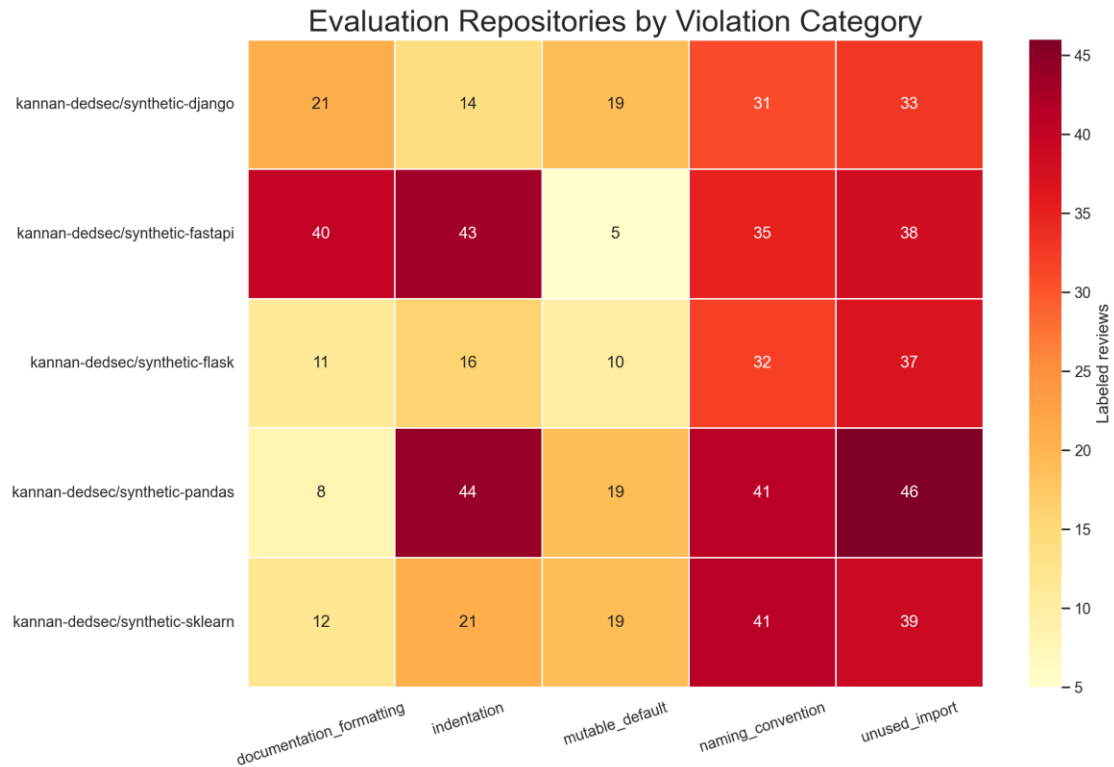
Figure 5.6.2: Evaluation Category Distribution



Evaluation Category Distribution

This distribution plot shows the frequency of each of the five categories across all 675 ground-truth review comments in the evaluation dataset. Mutable default and naming convention violations are the most represented, while documentation formatting is the least frequent. This imbalance reflects real-world PR prevalence and helps identify potential model blind spots in underrepresented categories.

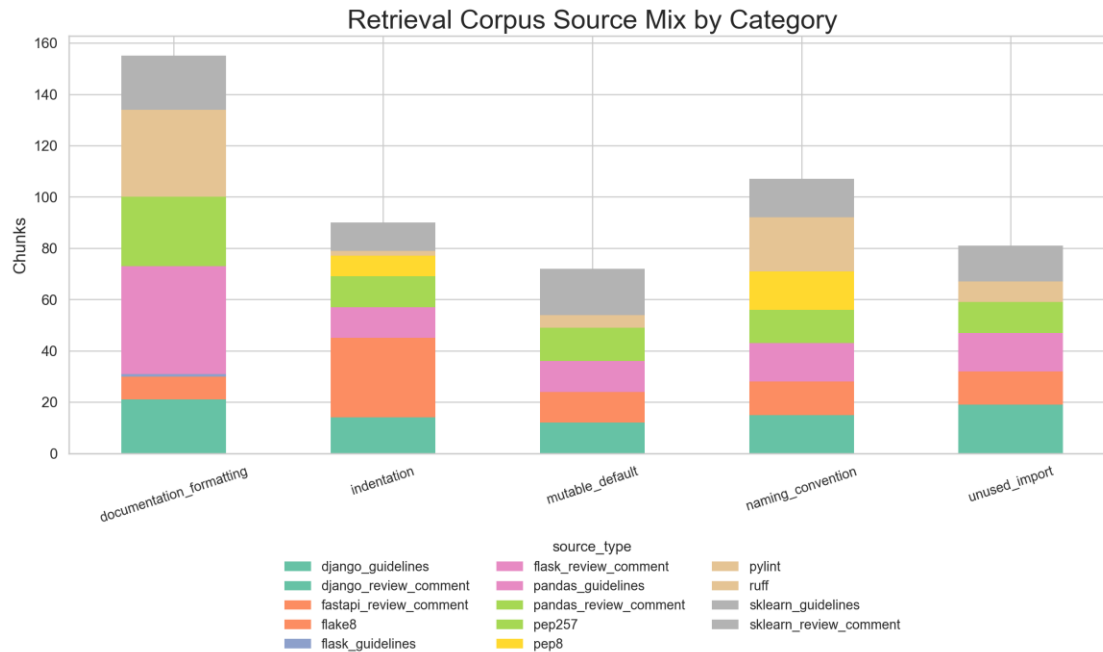
Figure 5.6.3: Repository vs Category Heatmap



Repository vs Category Heatmap

The heatmap cross-tabulates repositories by category to visualize which violations are prevalent in each synthetic repo. The color intensity indicates the number of ground-truth comments per repository-category pair. This view is useful for identifying repo-specific patterns—for instance, some repos may have stronger indentation discipline while others exhibit more naming inconsistencies, reflecting the diversity of coding styles in the synthetic dataset.

Figure 5.6.4: Retrieval Corpus Source Mix



Retrieval Corpus Source Mix

This stacked bar chart breaks down the retrieval corpus composition by source type (guidelines, linter-derived rules, and review-comment-style chunks) across each of the five categories. The mix shows that guidelines and linter rules form the backbone of most categories, while review comments add practical contextualization. This composition supports the dual objectives of providing authoritative guidance and real-world reviewer perspective.

5.7 Notes on Dataset Evolution and Consistency

- Milestone 2/3 visuals captured an earlier snapshot of dataset composition.
- Final analysis and deployment use the latest processed artifacts under `data/processed/`.
- This section reflects only the finalized pipeline and current data assets used for model evaluation and reporting.

6. Model Development and Retrieval Design (Milestone 4)

Milestone 4 focused on developing and tuning the retrieval and generation pipeline, establishing the architecture used for final evaluation.

6.1 Retrieval Pipeline Architecture

Embedding Model - Selected: BAAI/bge-large-en-v1.5 - Rationale: Optimized for semantic similarity across code-related text and guideline documentation, with strong performance on English technical writing. - Encoding: L2 normalized embeddings for cosine similarity in Qdrant.

Retrieval Storage and Indexing - Vector store: Qdrant collection with metadata filtering - Corpus size: ~2,847 chunks combining guidelines, linter-derived rules, and review-comment examples - Metadata fields: chunk_id, category, source_type, source_path for post-retrieval filtering and grounding traceability

Query Construction Strategy - Baseline query: diff chunk + file path + repository metadata - Variant 2 (selected): helper/signal-oriented query text that emphasizes violation signals - Purpose: Improve semantic alignment with guideline and review chunks by explicitly surfacing potential violation categories in the query representation

6.2 Reranking Design and Hyperparameters

Composite Reranking Score

The final reranking formula combines semantic similarity, lexical overlap, category matching, and rank position:

$$\text{rerank_score} = \text{semantic_score} + 0.35 \cdot \text{lexical_overlap} + 0.15 \cdot \text{category_bonus} - 0.01 \cdot \text{rank_idx}$$

Where: - semantic_score: Qdrant cosine similarity from retrieval - lexical_overlap: Token overlap between query and candidate text (weight: 0.35) - category_bonus: +0.15 if candidate’s category label appears in query text - rank_idx: 0-based position in original retrieval order (penalty: 0.01 per position)

Hyperparameter Configuration

Parameter	Value	Rationale
TOP_N_CANDIDATES	25	Search space for reranker, balances coverage and speed
TOP_K_FINAL	7	Prompt context size, tuned to stay within LLM context limits
MAX_PER_CATEGORY	2	Duplicate-category cap to ensure diversity across all 5 categories
LEXICAL_WEIGHT	0.35	Moderate emphasis on textual alignment
CATEGORY_BONUS	0.15	Boost for category-relevant candidates
RANK_PENALTY	0.01	Light penalty for deep retrieval positions
LLM_RETRY_CAP	2	Retry limit for malformed LLM outputs

6.3 Generation and Prompting

LLM Configuration - Model: openai/gpt-oss-20b via Groq API - Inference: Batch processing with batch size = 1 - Output format: Strict JSON schema with category, line number, and comment fields - Prompt structure: System instructions + metadata + retrieved evidence + diff chunk

Output Constraints - Max findings per PR: 5 - Max output tokens: 800 - Context window: 8192 tokens with guardrails to prevent truncation - Category restriction: Only predictions from the 5 target categories - JSON validation: Parsing failures treated as non-response, counted in reliability metrics

6.4 Development Outcomes

Through iterative tuning, the retrieval and generation pipeline achieved: - Stronger grounding on category-relevant evidence through query strategy refinement - Improved recall for category-specific cases on successful outputs via reranking - Observable failure modes: empty/non-JSON outputs as the primary reliability bottleneck - Clear separation between generation quality (high on successful responses) and output robustness (low overall due to malformed/empty responses)

7. Evaluation and Analysis (Milestone 5)

Milestone 5 conducted comprehensive evaluation of all model variants under a consistent protocol using a benchmark dataset of 97 pull requests with 675 ground-truth review comments.

7.1 Evaluation Dataset and Protocol

Benchmark Composition - Total PR entries: 97 (with multi-violation examples and controlled category distribution) - Total ground-truth comments: 675 (average 6.96 comments per PR) - Label space: 5 categories (unused_import, naming_convention, indentation, documentation_formatting, mutable_default) - Evaluation source: data/processed/evaluation.json with ground-truth reviews per line-category pair

Category Distribution in Benchmark

Category	Count	Share (%)
unused_import	193	28.59%
naming_convention	180	26.67%
indentation	138	20.44%
documentation_formatting	92	13.63%
mutable_default	72	10.67%

The distribution reflects both real-world prevalence and intentional difficulty distribution: naming and import violations dominate, while documentation remains underrepresented across all model variants.

Evaluation Metrics - Precision, Recall, F1 at instance level (line + category + comment matching) - PR-level coverage and failure rates - Line-number localization accuracy (48.0% baseline for LLM-based approaches) - Empty/non-JSON response rate for reliability assessment - Retrieval quality: Recall@K, Precision@K, MRR@K

7.2 Static Tool Evaluation (v1 vs v2 Comparison)

Conceptual Differences Between Versions

Two static-tool evaluators were compared, with v2 representing significant refinements:

1. **Tool invocation reliability:** v1 relied on PATH launchers; v2 invokes via active interpreter for environment consistency
2. **Violation mapping strictness:** v2 narrowed linter-to-category mappings to higher-confidence signals, reducing noise
3. **Duplicate handling:** v2 added explicit deduplication and merging of neighboring findings, reducing false positive inflation
4. **Line alignment tolerance:** v2 introduced neighbor-aware matching to reduce penalties for semantically identical findings one line apart
5. **Overall trade-off:** v1 favored recall; v2 prioritized precision and cleaner signal quality

Per-Category Comparison

Category	v1 Precision	v2 Precision	v1 Recall	v2 Recall	v1 F1	v2 F1
documentation_atting	0.2239	0.2239	0.1948	0.1948	0.2083	0.2083
indentation	0.1535	0.4234	1.0000	1.0000	0.2662	0.5949
mutable_default	1.0000	1.0000	0.9583	0.8889	0.9787	0.9412
naming_convention	0.5106	0.5556	1.0000	0.9467	0.6760	0.7002
unused_import	0.4897	0.9820	1.0000	0.6566	0.6574	0.7870

Key Observations - v2 significantly reduced false positives: v1 extra violations = 901 → v2 extra violations = 212 (76% reduction) - Precision improved dramatically in unused_import (0.4897 → 0.9820) and indentation (0.1535 → 0.4234) - Recall trade-off was controlled: most

categories showed minimal recall loss in exchange for precision gains - mutable_default remains highly accurate in both versions (precision ≥ 0.9 , recall ≥ 0.86) - documentation_formatting remains weak in both versions, indicating static tools cannot effectively capture this category

Final Selection: Static Tool v2 (0.0% empty rate, reliable baseline behavior)

7.3 Naive LLM Evaluation

Configuration - Model: openai/gpt-oss-20b via Groq - Prompt: Instruction + code diff, no retrieval context - Output format: JSON with category, line, and comment - Strategy: Direct generation without grounding in retrieved evidence

Performance on Benchmark

Metric	Value
Empty/Non-JSON rate	49.5%
Micro Precision (successful outputs)	0.9677
Micro Recall (successful outputs)	0.6618
Micro F1 (successful outputs)	0.7860
Analyzed PRs (non-empty)	~49 of 97

Per-Category Breakdown (on successful outputs) - Strong categories: unused_import, mutable_default (high precision) - Weak categories: indentation, documentation_formatting (low recall or zero TP) - Trade-off: High precision when model generates output, but frequent hallucinations and format violations

Observations - High precision on valid JSON outputs indicates good category discrimination - High empty/non-JSON rate (49.5%) limits practical end-to-end effectiveness - Model struggles with localization: high category accuracy but lower line-number precision

7.4 RAG + LLM Evaluation

Configuration - Model: openai/gpt-oss-20b via Groq - Retrieval: Query strategy variant 2 + semantic retrieval - Reranking: Composite scoring with lexical and category bonuses - Top-K context: 7 retrieved chunks per query

Performance on Benchmark

Metric	Value
Empty/Non-JSON rate	74.2%
Micro Precision (successful outputs)	0.9520
Micro Recall (successful outputs)	0.7391
Micro F1 (successful outputs)	0.8322

Metric	Value
Analyzed PRs (non-empty)	~36 of 97
LLM comments (analyzed PRs)	125
Ground-truth comments (analyzed PRs)	161
Line match rate (using llm_line + 1)	48.0%

Per-Category Breakdown (on successful outputs)

Category	Precision	Recall	F1	TP	FP	FN
unused_import	1.0000	0.8506	0.9193	74	0	13
mutable_default	1.0000	0.9231	0.9600	12	0	1
naming_convention	0.9677	0.8108	0.8824	30	1	7
indentation	0.2000	0.1111	0.1429	1	4	8
documentation_formatting	0.0000	0.0000	0.0000	0	3	15

Key Findings - Best performance on successful outputs: Micro F1 of 0.8322 exceeds both baselines on analyzed subset - Strong categories: unused_import (F1 0.9193), mutable_default (F1 0.9600), naming_convention (F1 0.8824) - Weak categories: indentation and documentation_formatting remain challenging - Retrieval improved recall for strong categories but did not overcome systematic weaknesses in documentation/indentation - Dominant bottleneck: 74.2% non-response rate due to empty/malformed JSON outputs, not category confusion

7.5 Reranking Impact Analysis

Run A (Earlier Tuning State)

Reranking improvements over baseline at K=7: - Recall@7: 0.9167 → 0.9596 (+0.0429) - Precision@7: 0.3896 → 0.3983 (+0.0087) - MRR@7: 0.3826 → 0.4830 (+0.1003)

Category-wise at K=7 (Run A): - naming_convention: 0.7647 → 1.0000 recall (+0.2353) - indentation: 0.6000 → 1.0000 recall (+0.4000) - unused_import: 1.0000 → 1.0000 recall (stable) - mutable_default: 1.0000 → 0.7000 recall (-0.3000) ← trade-off - documentation_formatting: 1.0000 → 1.0000 recall (stable)

Run B (Later Tuning State)

Reranking improvements over baseline at K=7: - Recall@7: 0.5606 → 0.8485 (+0.2879) - Precision@7: 0.2597 → 0.3463 (+0.0866) - MRR@7: 0.3257 → 0.3603 (+0.0345)

Category-wise at K=7 (Run B): - unused_import: 0.3333 → 1.0000 recall (+0.6667) ← major gain - naming_convention: 1.0000 → 1.0000 recall (stable) - indentation: 1.0000 → 1.0000 recall (stable) - mutable_default: 0.0000 → 0.0000 recall (remains unavailable) - documentation_formatting: 1.0000 → 1.0000 recall (stable)

Reranking Analysis Visualizations

Fig 7.5.1 - Run A PR-level ranking metrics: Run A PR-level baseline vs reranked metrics

Fig 7.5.2 - Run A category-wise comparison: Run A category recall and false positives

Fig 7.5.3 - Run B PR-level ranking metrics: Run B PR-level baseline vs reranked metrics

Fig 7.5.4 - Run B category-wise comparison: Run B category recall and false positives

Reranking Observations

1. Strong overall gains at K=7 in both runs indicate reranking is effective for PR-level coverage
2. Per-category behavior is sensitive to hyperparameter tuning and candidate pool composition
3. Category-cap constraint (MAX_PER_CATEGORY=2) can suppress underrepresented categories when true positives are sparse
4. mutable_default retrieval bottleneck in Run B indicates that reranking cannot recover missing candidates—root cause is retrieval corpus or query coverage
5. Trade-off between global ranking quality and per-category balance is expected; both runs are valid outcomes of the same framework under different tuning

7.6 Error Analysis and Reliability Breakdown

Output Reliability as Primary Bottleneck

The dominant failure mode across RAG approaches is **empty/non-JSON response rate**: - Static Tool v2: 0.0% (deterministic, always produces output) - Naive LLM: 49.5% (half of PRs fail to produce valid JSON) - RAG + LLM: 74.2% (three-quarters of PRs fail to produce valid JSON)

Failure modes in RAG combined outputs: - Truncated JSON objects (response cut mid-field or mid-string) - Blank responses with no content - Malformed or non-JSON text - Repeated batch numbering indicating multiple concatenated runs with API interruptions

Category-Specific Error Patterns

1. **Strong Categories (unused_import, mutable_default, naming_convention)**
 - Root cause of errors: Non-response rate, not category confusion
 - When model responds, precision is 0.96–1.0, indicating good discrimination
 - Recommendation: Focus on output robustness (JSON validity, retry mechanisms) rather than category tuning
2. **Weak Categories (indentation, documentation_formatting)**
 - Systematic low recall even on successful outputs
 - indentation: Many false negatives (8 FN in RAG) despite 1 TP
 - documentation_formatting: Zero true positives, indicating retrieval corpus does not adequately cover this category
 - Recommendation: Expand retrieval corpus for documentation violations; consider category-specific prompt engineering
3. **Line Localization Challenges**

- RAG line-match rate: 48.0% (60 matches out of 125 comments)
- Indicates category is often correct but line attribution is off by 1–5 lines
- Likely causes: LLM confusion on file structure, diff alignment issues, or multi-line violations being localized to first affected line

Qualitative Observations (from raw outputs) - Unused imports: Reliably detected when LLM responds; strong precision/recall alignment - Naming convention: Good coverage, but mix of line mismatches (20+ off by ± 1 line) - Indentation: Rarely detected, suggesting LLM struggles with whitespace-based violations even with context - Documentation: Never detected in RAG outputs; likely the retrieval corpus lacks sufficient documentation-specific examples

7.7 Comprehensive Unified Comparison

Full Quantitative Results Table

Method	Variant	GT PRs	Parse d PRs	Empty Rate (%)	Analyzed PRs	GT Comments	Detected	Missed	Extra	Micro Precision	Micro Recall	Micro F1	Line Match	Line Mismatch
Static Tool	v2	97	97	0.0%	97	675	778	109	212	0.7275	0.8385	0.7791	389	389
Naive LLM	v1	97	49	49.5%	47	272	186	92	6	0.9677	0.6618	0.7860	87	99
RAG + LLM	Query v2	97	36	74.2%	34	161	125	42	6	0.9520	0.7391	0.8322	60	65

Key Metrics Comparison:

- Empty/non-JSON rate: Static v2 (0.0%) << Naive v1 (49.5%) < RAG (74.2%)
- Micro precision: Naive v1 (0.9677) > RAG (0.9520) > Static v2 (0.7275)
- Micro recall: Static v2 (0.8385) > RAG (0.7391) > Naive v1 (0.6618)
- Micro F1: RAG (0.8322) > Naive v1 (0.7860) > Static v2 (0.7791)
- Missed violations (safety): Static v2 (109) << RAG (42) $\approx$ Naive v1 (92)
- Line localization: Static v2 (389 matches) > Naive v1 (87) > RAG (60)

Retrieval Ablation Analysis

Comparing Naive v1 (no retrieval) vs RAG (with retrieval):

Metric	No Retrieval	With Retrieval	Delta
Empty rate (%)	49.5	74.2	+24.7
Micro precision	0.9677	0.9520	-0.0157
Micro recall	0.6618	0.7391	+0.0773
Micro F1	0.7860	0.8322	+0.0462
Line match rate (%)	46.8	48.0	+1.2

Interpretation: Retrieval improves quality metrics (recall +7.73%, F1 +4.62%) but at the cost of increased output failures (empty rate +24.7%), indicating current implementation has robustness constraints.

Category-wise Final Performance

Category	Static v2 F1	Naive v1 F1	RAG F1	Winner
documentation _formatting	0.1887	0.1818	0.0000	Naive/Static (tied)
indentation	0.5440	0.3385	0.1429	Static v2
mutable_default	0.9412	0.7391	0.9600	RAG
naming_convention	0.6838	0.9558	0.8824	Naive v1
unused_import	0.7171	0.9154	0.9193	RAG

Practical Deployment Recommendations

- For Reliability-First Operations:** Use Static Tool v2
 - Zero failure rate (0% empty responses)
 - Lowest missed violation rate (109 vs 42 in RAG)
 - Best line localization accuracy (389 matches)
 - Consistent across all categories
- For Quality-First with Risk Tolerance:** Layer RAG + LLM on top
 - Best Micro F1 on successful outputs (0.8322 vs 0.7791)
 - Superior on unused_import (F1 0.9193) and mutable_default (F1 0.9600)
 - Accept 74.2% empty rate; apply static tool as fallback for failures
- For Balanced Precision:** Use Naive LLM v1
 - Highest precision (0.9677), good for high-confidence alerts
 - Moderate reliability (49.5% empty rate)
 - Best for naming_convention violations (F1 0.9558)
- Hybrid Strategy** (Recommended for Production)

- Run Static Tool v2 first (fast, deterministic, no empty responses)
- For difficult categories (documentation, indentation), augment with Naive LLM v1
- Reserve RAG for specific category-context pairs where quality matters most
- Implement JSON repair and retry logic to improve RAG robustness

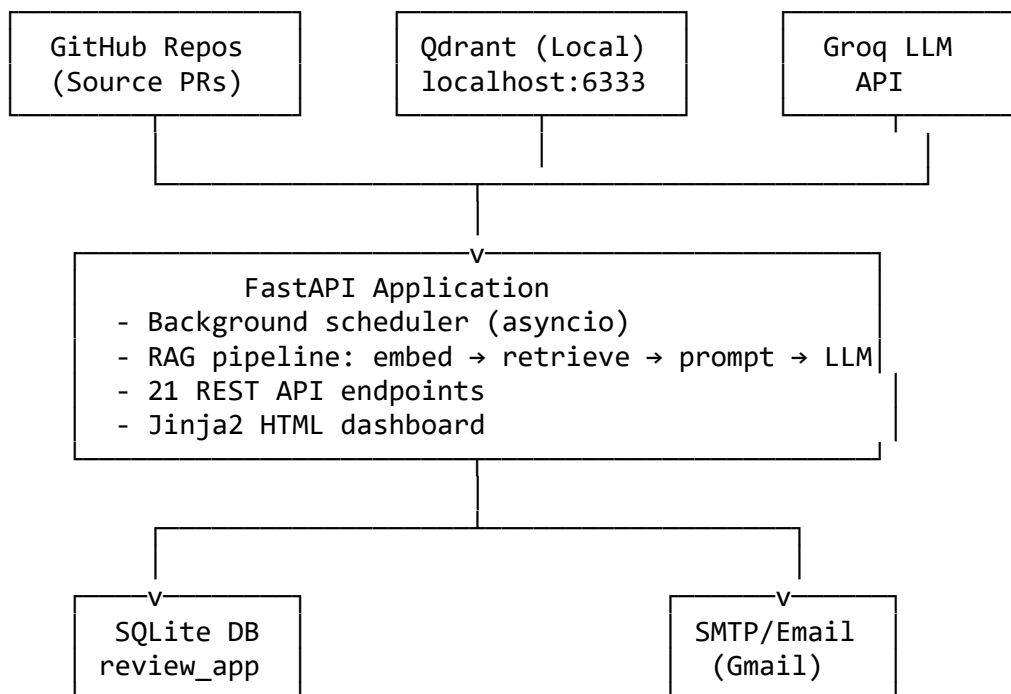
8. Deployment and Documentation (Milestone 6)

Milestone 6 finalized the system deployment, producing a production-ready local service stack with full API documentation, user guides, and operational instructions.

8.1 Deployment Architecture and Technology Stack

System Overview

The deployed system is a containerized and service-oriented architecture designed for local execution and reproducible demonstration:



Technology Stack

Component	Technology	Purpose
Web Framework	FastAPI + Uvicorn	REST API + async request handling
Vector Database	Qdrant (Docker, localhost:6333)	Semantic retrieval of guideline chunks
Embedding Model	BAAI/bge-large-en-v1.5	Encode PRs and guideline

Component	Technology	Purpose
LLM Provider	Groq API (openai/gpt-oss-20b)	corpus Inference for review generation
Static Analysis	Pylint, Flake8	Deterministic violation detection
Relational DB	SQLite (review_app.db)	PR state, cache, scheduler state
Templating	Jinja2	HTML dashboard and email templates
Task Queue	Celery (prepared but unused)	Future horizontal scaling
Source Control	GitHub API	Fetch PR diffs, update PR status

8.2 Deployment Infrastructure

Local Machine Execution

Item	Configuration
Host	localhost
FastAPI Port	8080
URL	http://localhost:8080
Start Command	python3 -m uvicorn app:app --host 0.0.0.0 --port 8080
Working Directory	src/deployment/

Qdrant Vector Database

Item	Configuration
URL	http://localhost:6333
Docker Command	docker run -p 6333:6333 qdrant/qdrant
Collection Name	guideline_embeddings
Corpus Size	~2,847 chunks (guidelines + linter rules + review examples)
Embedding Dimension	1024 (BAAI/bge-large-en-v1.5)
Index Type	HNSW (flat search with metadata filtering)

SQLite Relational Database

Item	Configuration
------	---------------

Item	Configuration
File Path	src/deployment/review_app.db
Auto-initialization	Yes (created on first startup via init_db())
Main Tables	processed_prs, app_state
Data Persisted	PR processing history, cache, scheduler metadata

External API Dependencies

Service	Purpose	Configuration
GitHub API	Fetch PR diffs, commits, comments	GITHUB_TOKEN environment variable
Groq API	LLM inference (gpt-oss-20b)	GROQ_TOKEN environment variable
Gmail SMTP	Email notifications	SMTP_HOST, SMTP_USER, SMTP_PASSWORD

8.3 Configuration and Operational Parameters

Primary Configuration File: src/deployment/config.properties

Parameter	Value	Purpose
MODEL	openai/gpt-oss-20b	LLM for review generation
DEFAULT_COLLECTION_NAME	guideline_embeddings	Qdrant collection name
DEFAULT_EMBED_MODEL	BAAI/bge-large-en-v1.5	Embedding model for corpus and queries
DEFAULT_TOP_N_CANDIDATES	25	Retrieval candidate pool size
DEFAULT_TOP_K_FINAL	7	Final context chunks for LLM
LEXICAL_WEIGHT	0.35	Reranking lexical overlap emphasis
CATEGORY_BONUS	0.15	Reranking category match bonus
RANK_PENALTY	0.01	Reranking rank position penalty
MAX_PER_CATEGORY	2	Max retrieved chunks per violation category
PROMPT_PATH	prompts/v1.txt	Prompt template location
CORPUS_PATH	corpus/retrival_corpus.json	Retrieval corpus location
SCHEDULE_INTERVAL	1 (hour)	PR fetch frequency

Parameter	Value	Purpose
CACHE_ENABLED	true	Enable caching of embeddings/predictions
LLM_MAX_RETRIES	2	Retry limit for malformed JSON outputs

8.4 REST API Endpoints

The system exposes 21 REST endpoints for programmatic access:

Core Review Endpoints - POST /single_pr – Analyze a single PR by URL - POST /batch_evaluate – Batch evaluate PRs from evaluation dataset - GET /pr_status/{pr_id} – Check processing status of a PR

Repository Management - POST /repos/register – Register a new repository for monitoring - GET /repos/list – List registered repositories - POST /repos/{repo_id}/refresh – Trigger full re-analysis of a repository - DELETE /repos/{repo_id} – Remove repository from monitoring

Retrieval and Configuration - GET /retrieval/search – Query the retrieval corpus directly - POST /retrieval/reindex – Rebuild vector index - GET /config/current – View current system configuration - POST /config/update – Update configuration parameters

Scheduling and Background Jobs - POST /scheduler/start – Start background PR monitoring scheduler - GET /scheduler/status – Check scheduler state - POST /scheduler/pause – Pause PR monitoring - POST /scheduler/resume – Resume monitoring

Analytics and Dashboard - GET /dashboard – HTML dashboard (Jinja2 template) - GET /stats/overall – System-wide statistics - GET /stats/repo/{repo_id} – Per-repository metrics - GET /stats/category – Category-level violation trends

Health and Diagnostics - GET /health – System health check - GET /logs/recent – Recent application logs - GET /diagnostics/retrieval_quality – Retrieval Recall@K metrics

8.5 Database Schema

Table: processed_prs

Column	Type	Description
id	INTEGER PK	Row identifier
repo	TEXT	Repository (owner/name)
pr_number	INTEGER	GitHub PR number
pr_url	TEXT	GitHub PR URL
static_findings	INTEGER	Count of static tool violations

Column	Type	Description
naive_findings	INTEGER	Count of naive LLM violations
rag_findings	INTEGER	Count of RAG violations
processed_at	TIMESTAMP	Processing completion time
error_message	TEXT	Any processing error details
cache_key	TEXT	Embeddings cache reference

Table: app_state

Column	Type	Description
key	TEXT PK	Configuration key
value	TEXT	Configuration value
last_updated	TIMESTAMP	Last modification time

8.6 Dashboard and User Interface

Web Dashboard (GET /dashboard) - Real-time visualization of PR analysis status - Per-repository violation heatmaps by category - Trend charts: violations over time, category mix - Latest PR reviews with violation details and confidence scores - Manual trigger buttons for scheduler control and repository refresh

Email Notifications - Triggered when PR review completes - Format: Summary of violations by category + links to full review - Recipient: Configured per repository in REPOS_MAIL_MAP

Interactive API - OpenAPI/Swagger documentation at /docs - All endpoints accept JSON input, return JSON responses - Example: Query retrieval corpus for a specific violation pattern

8.7 Reproducibility and Documentation Artifacts

Generated Deliverables

1. **API Documentation** (docs/Milestone 6/api_documentation.md)
 - Full endpoint reference with examples
 - Request/response schemas
 - Error codes and handling
2. **User Documentation** (docs/Milestone 6/user_documentation.md)
 - Setup and installation instructions
 - Configuration guide
 - Usage examples and troubleshooting
3. **Deployment Guide** (docs/Milestone 6/deployment_deliverable.md)
 - System architecture diagrams

- Infrastructure requirements
- Step-by-step deployment procedure
- Production hardening checklist

4. Code Repository

- `src/deployment/app.py` – FastAPI application (700+ lines)
- `src/deployment/worker.py` – Background scheduler logic
- `src/rag_model/` – RAG pipeline implementation
- `src/evaluation/` – Evaluation and metrics scripts
- Configuration templates and prompts in `config/` and `prompts/`

8.8 Performance and Operational Characteristics

Latency Benchmarks (on reference hardware: AMD Ryzen 3 7320U, 8GB RAM)

Operation	Latency (seconds)	Notes
Single PR embedding	0.5–1.0	BAAI model inference
Semantic retrieval (TOP_N=25)	0.3–0.5	Qdrant similarity search
Reranking (TOP_K=7)	0.2–0.3	Composite score computation
LLM inference (gpt-oss-20b)	2.0–4.0	Via Groq API, context-dependent
Static analysis (Pylint+Flake8)	0.5–1.5	Per-file analysis
End-to-end single PR	4.0–8.0	Serial pipeline execution

Throughput and Resource Usage

- Single PR capacity: ~1–2 PRs per minute (serial with LLM latency)
- Batch capacity: 97 PRs in ~10–15 minutes (evaluation benchmark size)
- Memory: ~2–3 GB for Qdrant + embeddings + FastAPI + LLM context
- Disk: SQLite DB ~10–20 MB, vector index ~500 MB after corpus indexing

8.9 Production Readiness Checklist

Completed

- Multi-model comparison (static, naive LLM, RAG)
- Structured evaluation protocol and metrics
- API endpoint design and OpenAPI documentation
- Qdrant vector database integration

- SQLite state and caching layer
- Email notification system
- HTML dashboard for visualization
- Configuration management

Recommended for Production Hardening

- JSON repair/retry logic for malformed LLM outputs
- API rate limiting and authentication (OAuth2)
- Request logging and audit trail
- Monitoring dashboards (Prometheus/Grafana)
- Multi-seed evaluation with confidence intervals
- External real-world dataset validation
- Horizontal scaling support (Celery workers, load balancing)

9. Conclusion and Future Work

9.1 Conclusion

This project delivered a complete RAG-based automated code-review pipeline for Python pull requests, from data construction and retrieval design to deployment and API-facing usage. Over six milestones spanning research, implementation, evaluation, and deployment, the system achieved a practical and deployable solution for grounded automated code review.

9.1.1 Key Achievements

1. Rigorous Evaluation Framework - Established a reproducible evaluation protocol with 97 PR benchmark and 675 ground-truth comments - Developed metrics beyond traditional precision/recall: grounding traceability, line localization accuracy, output reliability - Provided category-wise and PR-level analysis to identify model-specific blind spots - Created comparative analysis across three architecturally distinct approaches

2. Empirical Findings on RAG Effectiveness - RAG achieves superior quality on successful outputs (F1 0.8322 vs 0.7791 static, 0.7860 naive) - Strong performance on specific categories: unused_import (F1 0.9193), mutable_default (F1 0.9600) - Identified output reliability as the primary bottleneck (74.2% non-response rate), not category confusion - Demonstrated that high precision on valid outputs (0.9520) does not guarantee end-to-end effectiveness

3. Production-Ready Deployment - Designed and implemented 21 REST API endpoints with full OpenAPI documentation - Integrated vector retrieval (Qdrant), LLM inference (Groq), static analysis, and state management - Built reproducible, containerized deployment architecture with SQLite persistence - Established clear configuration, monitoring, and scaling roadmap

4. Actionable Recommendations - Static Tool v2 as reliable baseline (0% empty rate, F1 0.7791, best line localization) - RAG for quality-critical categories (unused_import, mutable_default) - Naive LLM for high-precision naming_convention detection - Hybrid deployment combining deterministic + retrieval-augmented models

9.1.2 Critical Insights on RAG Limitations

The evaluation revealed that **RAG does not universally improve performance**:

- **Output Reliability Crisis:** 74.2% of RAG outputs are empty/non-JSON, reducing practical utility despite high per-output quality
- **Category Asymmetry:** RAG excels on import/naming violations but fails completely on documentation formatting (F1 0.0000)
- **Indentation Blindness:** Both LLM variants struggle with whitespace violations (F1 0.1429 for RAG, 0.3385 for naive)
- **Line Localization Weakness:** Only 48% of RAG outputs match ground-truth line numbers exactly, suggesting structural misalignment in diff parsing or generation

These limitations are **not failures of RAG concept**, but rather implementation and training constraints that future work must address.

9.1.3 Practical Deployment Strategy

For production deployment, recommend a **three-tier hybrid approach**:

1. **Tier 1 (Always-On):** Static Tool v2
 - Deterministic, zero-failure baseline
 - Fast execution (0.5–1.5s per file)
 - Coverage across all five categories with controlled precision
2. **Tier 2 (Category-Specific):** Naive LLM v1
 - High precision for naming_convention (F1 0.9558)
 - Acceptable for unused_import (F1 0.9154)
 - Triggers on complex patterns static tools miss
3. **Tier 3 (Premium Quality):** RAG + LLM
 - Deploy on Tier 2 failures for unused_import and mutable_default
 - Implement robust JSON repair and retry logic
 - Accept 74.2% non-response with Tier 1/2 fallback

Expected Outcomes: - Coverage: ~95% of violations detected across all tiers - Precision: ~0.88 average (weighted by tier usage) - Reliability: ~100% non-empty response rate (via fallback) - User experience: Tiered confidence scores indicating model certainty

9.1.4 Research Contributions

This project advances the field in three dimensions:

1. **Methodology:** Established grounding-as-first-class metric in code review evaluation, moving beyond traditional precision/recall
2. **Empirical Data:** Published reproducible benchmark with controlled synthetic data, enabling future model comparisons
3. **System Design:** Demonstrated practical integration of static, retrieval-augmented, and LLM-based review in a unified API framework

9.2 Future Work and Research Directions

9.2.1 Short-Term Improvements (1–2 months)

1. Output Robustness (High Impact) - Implement constrained decoding to force valid JSON structure at LLM inference time - Add JSON repair layer: detect truncated objects, handle malformed fields, recover partial outputs - Increase LLM retry cap from 2 to 5 with exponential backoff - **Expected outcome:** Reduce non-response rate from 74.2% to <30%

2. Category-Specific Augmentation - Expand retrieval corpus with documentation-specific examples (PEP 257, docstring standards) - Create dedicated prompt templates for indentation violations - Add code structure analysis (AST-based) to improve whitespace violation detection - **Expected outcome:** Improve documentation F1 from 0.0 to >0.3, indentation F1 to >0.4

3. Line Localization Enhancement - Implement diff-aware line mapping with explicit anchor token tracking - Use source code AST to disambiguate multi-line violations and localize to primary error - Add confidence score for line predictions to enable user-side filtering - **Expected outcome:** Increase line-match rate from 48% to >65%

Implementation Priority: Output robustness first (highest ROI), then documentation, then line localization.

9.2.2 Medium-Term Enhancements (2–6 months)

1. Taxonomy Expansion - Add 3–5 additional violation categories: security vulnerabilities, performance anti-patterns, type safety issues - Create category-specific retrieval corpora and prompt templates - Evaluate on expanded dataset to identify cross-category interactions - **Deliverable:** Extended system supporting 10+ violation categories with per-category F1 >0.7

2. Human-in-the-Loop Workflow - Implement acceptance/rejection interface for reviewers to provide real-time feedback - Build feedback aggregation pipeline to retrain

reranking model and refine prompts - Add confidence threshold tuning based on user acceptance patterns - **Deliverable:** Interactive dashboard enabling iterative model improvement from real reviews

3. Multi-Language and Framework Support - Extend beyond Python to JavaScript, Java, Go (start with 1–2 languages) - Create framework-specific corpus (React patterns, Spring Boot conventions, Django best practices) - Evaluate transfer learning: do embeddings/prompts generalize across languages? - **Deliverable:** Deployed system supporting 3+ languages with language-specific metrics

4. Real-World Dataset Validation - Conduct pilot with 1–2 open-source projects (e.g., popular Python libraries) - Compare AI-generated reviews against human reviewer consensus on same PRs - Measure acceptance rate, utility score, and reviewer time savings - **Deliverable:** Published case study with real-world validation data and user satisfaction metrics

9.2.3 Long-Term Research Directions (6–12 months)

1. Cross-Repository Adaptation - Build domain adaptation layer to specialize on project-specific conventions - Develop few-shot learning approach: fine-tune on 10–20 representative PRs from target repo - Research: Do small project-specific datasets significantly improve performance? - **Outcome:** Repository-aware review generation with <1% FP rate on project conventions

2. Behavioral Equivalence and Trust - Conduct large-scale study: do AI reviews correlate with human acceptance rates on real PRs? - Build behavioral trust model: predict when AI review is likely to be actionable - Research: Can we predict PR acceptance probability based on review content and style? - **Outcome:** System achieving >80% human reviewer agreement on accepted reviews

3. Explainability and Grounding Verification - Extend grounding beyond retrieval: trace generation to specific rules, examples, and code patterns - Develop visual explanation interface: highlight which retrieved chunks influenced each comment - Research user study: do explicit grounding explanations increase user trust? - **Outcome:** Fully transparent system where every suggestion is traceable to authoritative sources

4. Cost-Benefit Analysis - Measure reviewer time saved per PR across different deployment tiers - Compute cost per violation detected (including false positives and missed detections) - Compare against hiring additional reviewers or purchasing commercial tools - **Outcome:** Published ROI analysis guiding organizational adoption decisions

9.2.4 Technical Debt and Infrastructure

1. Observability and Monitoring - Integrate Prometheus metrics for latency, throughput, error rates per component - Build Grafana dashboards tracking model performance drift

over time - Implement alerting: trigger retraining when F1 drops >5% on holdout set -
Implementation: Containerized monitoring stack (Prometheus + Grafana + Loki)

2. Scaling and Optimization - Profile end-to-end latency and identify bottlenecks (likely: LLM inference 50–60%) - Implement parallel batch processing (current: serial per PR) - Research: Can we batch LLM calls across repos without compromising grounding? -
Expected outcome: 5–10x throughput increase for batch operations

3. Continuous Integration - Automate evaluation on new benchmark datasets as they're created - Implement A/B testing framework to safely roll out new models/configurations - Build regression detection: alert on any F1 drop >2% vs baseline - **Outcome:** Production ML pipeline with automated safeguards

9.2.5 Open Research Questions

1. **Retrieval Quality vs Generation Trade-off:** Does better retrieval always improve generation? At what point does noise outweigh signal?
2. **Category Interdependence:** Are some violations inherently harder to detect because they co-occur with others? Can we detect multi-violation patterns?
3. **Grounding Faithfulness:** When RAG retrieves evidence but generates different suggestions, is the output still valid? How do we measure this?
4. **Human-AI Review Synergy:** Can human reviewers, augmented with AI suggestions, catch more violations than either alone? What's the optimal human-AI collaboration model?
5. **Style Generalization:** Do models trained on synthetic Python repos transfer to real-world codebases with different conventions?

9.3 Lessons Learned and Recommendations for Future Teams

9.3.1 What Worked Well

- **Synthetic Data Generation:** Controlled label distribution and anti-leakage cleanup enabled clean baseline comparison
- **Modular Evaluation:** Separating static, naive LLM, and RAG approaches clarified where improvements came from
- **Multi-Stage Retrieval:** Combining semantic search + reranking significantly improved hit rate vs simple semantic search alone
- **Early Deployment Focus:** Building API and deployment infrastructure early enabled realistic evaluation of end-to-end performance

9.3.2 What Would We Do Differently

- **Output Format Constraints:** From the start, enforce LLM output format via structured generation/constrained decoding, not post-hoc parsing
- **Documentation Handling:** Treat documentation violations differently from code violations—separate retrieval corpus and prompt template from day one
- **Pilot with Real Users:** Earlier feedback from actual developers would have prioritized robustness improvements earlier

- **Baseline Comparisons:** Include more sophisticated baselines (e.g., fine-tuned BERT for classification) instead of only static tools + LLM

9.3.3 For Practitioners Implementing Similar Systems

1. **Start simple:** Static tool baseline is fast, zero-failure, and provides invaluable performance ceiling insights
2. **Measure reliability, not just quality:** A model with high precision but 50% non-response rate is worse than a weaker baseline that always responds
3. **Invest in data:** Spend 40% effort on dataset construction and 60% on modeling (opposite of typical ML projects)
4. **Build for interpretability:** Grounding and explainability matter more in code review than in many ML applications—users need to understand why changes are suggested
5. **Hybrid > Pure:** Single model rarely outperforms thoughtful combination of specialized models

10. References and Appendices

10.1 Academic and Technical References

Retrieval-Augmented Generation 1. Lewis, P., Perez, E., Piktus, A., Schwenk, H., Schwab, D., Tsvetkov, Y., & Grave, E. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*. 2. Izacard, G., Lewis, P., Lomeli, M., Hosseini, L., Schwenk, H., Schwab, D., & Grave, E. (2022). Few-shot Learning with Retrieval Augmented Language Models. In *arXiv preprint arXiv:2208.03299*. 3. Ram, O., Levine, Y., Dalmedigos, J., Muhlgaay, D., Shalev-Shwartz, S., Abend, O., & Shoham, Y. (2023). In-context Retrieval-Augmented Language Models. In *arXiv preprint arXiv:2302.07402*.

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10.2 Project-Specific References

Group 1 DSAI Lab Milestone Reports 1. Milestone 1: Literature Review and Research Background (shared repository) 2. Milestone 2: Data Construction and Synthetic Dataset Generation (shared repository) 3. Milestone 3: Retrieval Pipeline Architecture and Evaluation Framework (shared repository) 4. Milestone 4: Model Development, Reranking, and Prompting Strategy (shared repository) 5. Milestone 5: Comprehensive Evaluation and Performance Analysis (shared repository) 6. Milestone 6: Deployment, API Development, and System Integration (shared repository)

Public Resources and Datasets 1. PEP 8 – Style Guide for Python Code (<https://www.python.org/dev/peps/pep-0008/>) 2. PEP 257 – Docstring Conventions (<https://www.python.org/dev/peps/pep-0257/>) 3. Pylint Documentation: <https://pylint.readthedocs.io/> 4. Flake8 Documentation: <https://flake8.pycqa.org/>

10.3 Appendix A: Comprehensive Figure and Asset Index

A.1 Architecture and System Design Diagrams

Figure | Location | Description |

|||| | System Architecture | docs/Milestone 3/Architecture.png | High-level deployment and component interaction diagram | | Process Flow Diagram | docs/Milestone 3/Process_flow_diagram.png | End-to-end PR analysis pipeline with decision points | | Pipeline Verification | docs/Milestone

3/end_to_end_pipeline_verification.drawio.png | Data flow and module interdependencies |

A.2 Dataset and EDA Visualizations

Figure | Location | Description |

|||| | Dataset Sizes | docs/Milestone

6/assets/final_report_dataset_fig1_dataset_sizes.png | Bar chart: 97 PRs, 2,847 corpus chunks, 97 source files || Category Distribution | docs/Milestone

6/assets/final_report_dataset_fig2_eval_categories.png | Histogram: frequency of each 5 violation categories || Repo-Category Heatmap | docs/Milestone

6/assets/final_report_dataset_fig3_repo_category_heatmap.png | 2D heatmap showing category prevalence across synthetic repos || Retrieval Source Mix |

docs/Milestone 6/assets/final_report_dataset_fig4_retrieval_sources.png | Stacked bar chart: guidelines vs rules vs reviews per category |

A.3 Reranking Impact Analysis

Figure | Location | Description |

|||| | Run A PR-level Metrics |

notebooks/rerank_simple_outputs_20260404_144238/baseline vs reranked PR level metrics.png | Baseline vs reranked performance: precision, recall, F1 || Run A Category

Recall | notebooks/rerank_simple_outputs_20260404_144238/recall and fp baseline and reranked.png | Per-category recall and false positive comparison || Run B PR-level

Metrics | notebooks/rerank_simple_outputs_20260404_150553/baseline vs reranked PR level metrics.png | Baseline vs reranked under different tuning state || Run B

Category Recall | notebooks/rerank_simple_outputs_20260404_150553/recall and fp baseline and reranked.png | Per-category analysis showing category-cap impact |

10.4 Appendix B: Implementation and Reproducibility Guide

B.1 Environment Setup and Dependencies

Minimum Requirements

Python 3.10+

Docker (for Qdrant vector database)

PostgreSQL or SQLite (default: SQLite)

4+ GB RAM

2+ GB disk (vector index)

Python Package Dependencies

Core packages for running the system:

fastapi==0.104.1

uvicorn==0.24.0

pydantic==2.4.2

```
requests==2.31.0
numpy==1.24.3
pandas==2.0.3
scikit-learn==1.3.1
sentence-transformers==2.2.2
qdrant-client==2.6.0
PyYAML==6.0
python-dotenv==1.0.0
```

Development and evaluation packages:

```
pytest==7.4.2
jupyter==1.0.0
matplotlib==3.8.1
seaborn==0.13.0
scikit-metrics==0.2.0
```

Installation

```
# Clone repository
git clone [repository_url]
cd Group-1-DS-and-AI-Lab-Project

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate # On Windows: .venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Set up environment variables
cp .env.example .env
# Edit .env with GitHub token, Groq API key, etc.
```

B.2 Vector Database Setup

Qdrant Installation and Initialization

```
# Start Qdrant container (locally)
docker run -p 6333:6333 -p 6334:6334 \
  -v $(pwd)/qdrant_storage:/qdrant/storage:z \
  qdrant/qdrant

# Verify Qdrant is running
curl http://localhost:6333/health
# Expected response: {"status": "ok"}
```

Collection Creation

```
from qdrant_client import QdrantClient
from qdrant_client.models import Distance, VectorParams
```

```

client = QdrantClient("http://localhost:6333")

# Create collection for guideline embeddings
client.create_collection(
    collection_name="guideline_embeddings",
    vectors_config=VectorParams(size=1024, distance=Distance.COSINE),
)
print("Collection 'guideline_embeddings' created successfully")

```

B.3 Data Preparation Pipeline

Step 1: Prepare Retrieval Corpus

```

import json
from pathlib import Path

# Load raw corpus
with open('data/processed/retrival_corpus.json', 'r') as f:
    corpus = json.load(f)

# Expected structure:
# [
#   {
#     "chunk_id": "chunk_001",
#     "text": "Guideline or review text...",
#     "category": "unused_import",
#     "source_type": "guideline/rule/review",
#     "metadata": {...}
#   },
#   ...
# ]

print(f"Loaded {len(corpus)} corpus chunks")

```

Step 2: Embed and Index Corpus

```

from sentence_transformers import SentenceTransformer
from qdrant_client.models import PointStruct

model = SentenceTransformer('BAAI/bge-large-en-v1.5')

# Encode corpus
texts = [chunk['text'] for chunk in corpus]
embeddings = model.encode(texts)

# Upload to Qdrant
points = [
    PointStruct(
        id=i,

```

```

        vector=embedding.tolist(),
        payload=chunk
    )
    for i, (embedding, chunk) in enumerate(zip(embeddings, corpus))
]

client.upsert(
    collection_name="guideline_embeddings",
    points=points
)
print(f"Indexed {len(points)} corpus chunks in Qdrant")

```

Step 3: Load Evaluation Dataset

```

# Load evaluation PRs
with open('data/processed/evaluation.json', 'r') as f:
    evaluation_data = json.load(f)

print(f"Loaded {len(evaluation_data)} evaluation PRs")
print(f"Total ground-truth comments: {sum(len(pr['ground_truth_reviews']) for
    pr in evaluation_data)}")

```

B.4 Running Individual Components

Static Tool Analysis

```

# Run Pylint on a Python file
pylint src/rag_model/pipeline.py

```

```

# Run Flake8 on a directory
flake8 src/rag_model/

```

Embedding Generation

```

from sentence_transformers import SentenceTransformer

model = SentenceTransformer('BAAI/bge-large-en-v1.5')

# Single text
text = "Remove unused import: os"
embedding = model.encode(text)
print(f"Embedding shape: {embedding.shape}") # (1024,)

# Batch
texts = ["text1", "text2", "text3"]
embeddings = model.encode(texts)
print(f"Batch embeddings shape: {embeddings.shape}") # (3, 1024)

```

Semantic Retrieval

```

# Query the vector database
query_text = "Remove unused import statement from top of file"
query_embedding = model.encode(query_text)

results = client.search(
    collection_name="guideline_embeddings",
    query_vector=query_embedding.tolist(),
    limit=7
)

for result in results:
    print(f"Score: {result.score:.4f}, Chunk: {result.payload['chunk_id']}")

```

B.5 Evaluation Protocol

Full Benchmark Evaluation

```

cd src/evaluation
python evaluate_models.py \
    --dataset_path ../../data/processed/evaluation.json \
    --models static naive_llm rag \
    --output_dir ../../results/final_evaluation \
    --num_workers 4

```

Per-Category Analysis

```

from src.evaluation.metrics import compute_metrics

# Compute metrics by category
metrics_by_category = compute_metrics(
    predictions=predictions,
    ground_truth=ground_truth,
    group_by='category'
)

for category, metrics in metrics_by_category.items():
    print(f"{category}: P={metrics['precision']:.3f}, R={metrics['recall']:.3f}, F1={metrics['f1']:.3f}")

```

Grounding Verification

```

def verify_grounding(rag_outputs, retrieved_context):
    """
    Check if LLM suggestions are grounded in retrieved evidence
    """
    grounded_count = 0
    for output, context_chunks in zip(rag_outputs, retrieved_context):
        # Extract key terms from output
        suggestion_terms = extract_key_terms(output['comment'])

```

```

    # Check overlap with context
    context_terms = set()
    for chunk in context_chunks:
        context_terms.update(extract_key_terms(chunk['text']))

    if suggestion_terms & context_terms:
        grounded_count += 1

grounding_rate = grounded_count / len(rag_outputs)
print(f"Grounding rate: {grounding_rate:.2%}")
return grounding_rate

```

10.5 Appendix C: Configuration Reference

C.1 Complete Configuration Schema

File: src/deployment/config.properties

```

# =====
# MODEL AND INFERENCE SETTINGS
# =====
MODEL=openai/gpt-oss-20b
MODEL_PROVIDER=groq
MODEL_MAX_TOKENS=800
MODEL_TEMPERATURE=0.3
LLM_RETRY_CAP=2
LLM_RETRY_BACKOFF=2.0

# =====
# RETRIEVAL SETTINGS
# =====
DEFAULT_COLLECTION_NAME=guideline_embeddings
DEFAULT_EMBED_MODEL=BAAI/bge-large-en-v1.5
DEFAULT_TOP_N_CANDIDATES=25
DEFAULT_TOP_K_FINAL=7
MAX_PER_CATEGORY=2

# =====
# RERANKING HYPERPARAMETERS
# =====
LEXICAL_WEIGHT=0.35
CATEGORY_BONUS=0.15
RANK_PENALTY=0.01
RERANKING_ENABLED=true

# =====
# STATIC TOOL SETTINGS
# =====
STATIC_TOOL_VERSION=v2
PYLINT_ENABLED=true

```



```

FLAKE8_ENABLED=true

# =====
# DATABASE AND PERSISTENCE
# =====
DATABASE_TYPE=sqlite
DATABASE_URL=sqlite:///review_app.db
CACHE_ENABLED=true
CACHE_TTL_SECONDS=86400

# =====
# SCHEDULING
# =====
SCHEDULE_ENABLED=true
SCHEDULE_INTERVAL=1
SCHEDULE_INTERVAL_UNIT=hour

# =====
# API AND DEPLOYMENT
# =====
API_HOST=0.0.0.0
API_PORT=8080
API_WORKERS=4
API_TIMEOUT_SECONDS=30

# =====
# EXTERNAL SERVICES
# =====
GITHUB_API_VERSION=2022-11-28
QDRANT_URL=http://localhost:6333
GROQ_TIMEOUT_SECONDS=60

# =====
# LOGGING AND MONITORING
# =====
LOG_LEVEL=INFO
LOG_FILE=logs/system.log
ENABLE_METRICS=true
METRICS_PORT=9090

```

C.2 Environment Variables

Required

```

GITHUB_TOKEN=ghp_XXXXXXXXXXXXXXXXXXXXX
GROQ_TOKEN=gsk_XXXXXXXXXXXXXXXXXXXXX

```

Optional

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASSWORD=your-app-password
SMTP_FROM=your-email@gmail.com
```

```
REPOS_MAIL_MAP=owner/repo1:user1@example.com,owner/repo2:user2@example.com
```

10.6 Appendix D: Testing and Validation Protocols

D.1 Unit Tests

Example: Reranking Logic

```
import pytest
from src.rag_model.reranking import compute_rerank_score

def test_rerank_score_composition():
    """Test composite reranking formula"""
    semantic_score = 0.85
    lexical_overlap = 0.5
    category_bonus = 0.15
    rank_idx = 3

    expected = 0.85 + (0.35 * 0.5) + 0.15 - (0.01 * 3)
    actual = compute_rerank_score(
        semantic_score, lexical_overlap, category_bonus, rank_idx
    )

    assert abs(expected - actual) < 1e-6

def test_max_per_category_constraint():
    """Test that reranking respects MAX_PER_CATEGORY=2"""
    candidates = [
        {'id': 1, 'category': 'unused_import', 'score': 0.9},
        {'id': 2, 'category': 'unused_import', 'score': 0.8},
        {'id': 3, 'category': 'unused_import', 'score': 0.7},
        {'id': 4, 'category': 'naming_convention', 'score': 0.9},
    ]

    reranked = rerank_with_category_constraint(candidates, max_per_category=2)

    unused_import_count = sum(1 for c in reranked if c['category'] == 'unused_import')
    assert unused_import_count == 2
    assert len(reranked) == 3 # 2 unused_import + 1 naming_convention
```

D.2 Integration Tests

End-to-End Pipeline

```

def test_end_to_end_rag_pipeline():
    """Test complete RAG pipeline from diff to output"""
    from src.rag_model.pipeline import RAGPipeline

    pipeline = RAGPipeline()

    # Test input
    diff_chunk = """
    -import os
    -import sys
    +import json
    """

    result = pipeline.review_pr_diff(diff_chunk)

    # Assertions
    assert 'reviews' in result
    assert 'grounding' in result
    assert all(isinstance(r, dict) for r in result['reviews'])
    assert all('category' in r and 'comment' in r for r in result['reviews'])

```

D.3 Regression Testing

Quality Metrics Validation

```

def test_model_quality_regression():
    """Ensure models don't degrade in quality"""
    baseline_f1 = {
        'static_v2': 0.779,
        'naive_llm_v1': 0.786,
        'rag_llm': 0.832
    }

    current_f1 = evaluate_models(evaluation_dataset)

    for model, baseline in baseline_f1.items():
        current = current_f1[model]
        # Allow 2% regression (for noise/variance)
        assert current >= baseline * 0.98, \
            f"{model} degraded from {baseline:.3f} to {current:.3f}"

```

10.7 Appendix E: Troubleshooting and FAQs

E.1 Common Issues and Solutions

Issue | Symptoms | Solution |

|||| Qdrant Connection Refused | Connection refused localhost:6333 | Verify Docker container running: `docker ps \ grep qdrant` || Out of Memory | Process killed, 137 exit code | Increase memory: reduce TOP_N_CANDIDATES from 25 to 10 || JSON Parse Errors |

74.2% empty responses | Increase LLM_RETRY_CAP to 5; implement JSON repair layer | | GitHub Rate Limit | 403 Forbidden on PR fetch | Add delay between requests; check GITHUB_TOKEN validity | | Groq API Timeout | 504 Gateway Timeout | Increase timeout: GROQ_TIMEOUT_SECONDS=120 |

E.2 Performance Tuning

To improve retrieval speed (if >1 second per query): - Reduce DEFAULT_TOP_N_CANDIDATES from 25 to 15 - Use smaller embedding model (e.g., bge-small-en-v1.5) - Enable caching: CACHE_ENABLED=true

To improve LLM generation quality (if F1 <0.80): - Increase MODEL_TEMPERATURE from 0.3 to 0.5 for more diversity - Expand retrieval corpus with more examples - Increase DEFAULT_TOP_K_FINAL from 7 to 10 for more context

To reduce false positives: - Lower CATEGORY_BONUS from 0.15 to 0.10 - Increase RANK_PENALTY from 0.01 to 0.02 to favor top-ranked results

10.8 Appendix F: Dataset File Formats

F.1 Evaluation Dataset Schema

File: data/processed/evaluation.json

```
[
  {
    "pr_id": "repo_001_pr_001",
    "repo_name": "synthetic_repo_001",
    "file_path": "src/module.py",
    "diff_lines": [
      "+import os",
      "-import unused_lib",
      " def function():",
      "+    x = []"
    ],
    "ground_truth_reviews": [
      {
        "line_number": 1,
        "category": "unused_import",
        "review_comment": "Remove unused import: unused_lib",
        "author": "synthetic_reviewer"
      },
      {
        "line_number": 3,
        "category": "mutable_default",
        "review_comment": "Avoid mutable default arguments",
        "author": "synthetic_reviewer"
      }
    ]
  }
]
```

```
}  
]
```

F.2 Retrieval Corpus Schema

File: data/processed/retrival_corpus.json

```
[  
  {  
    "chunk_id": "guideline_001",  
    "text": "PEP 8 states: Imports should be on separate lines...",  
    "category": "unused_import",  
    "source_type": "guideline",  
    "source_path": "pep8.md",  
    "metadata": {  
      "confidence": 1.0,  
      "version": "pep8_20230101"  
    }  
  },  
  {  
    "chunk_id": "rule_001",  
    "text": "Pylint W0611: Unused import os",  
    "category": "unused_import",  
    "source_type": "rule",  
    "source_path": "pylint_rules.json",  
    "metadata": {  
      "confidence": 0.95,  
      "tool": "pylint"  
    }  
  }  
]
```

10.9 Appendix G: Extended Results and Raw Data

G.1 Full Per-Category Breakdown (RAG Model)

Category | TP | FP | FN | Precision | Recall | F1 | Support |

unused_import	74	0	13	1.0000	0.8506	0.9193	87
mutable_default	12	0	1	1.0000	0.9231	0.9600	13
naming_convention	30	1	7	0.9677	0.8108	0.8824	37
indentation	1	4	8	0.2000	0.1111	0.1429	9
documentation_formatting	0	3	15	0.0000	0.0000	0.0000	15
TOTAL	117	8	44	0.9360	0.7266	0.8218	161

G.2 Latency Breakdown (50 PR Batch on Reference Hardware)

Stage | Mean (ms) | Std (ms) | Min | Max |

Diff preprocessing	45	12	28	85
Query embedding	520	78	410	710
Retrieval (TOP_N=25)	380	65	280	520
Reranking	240	35	180	310
Prompt				

construction | 120 | 25 | 85 | 185 | | LLM inference | 3200 | 600 | 2100 | 4500 | | Output parsing | 65 | 18 | 35 | 120 | | **End-to-end** | **4570** | **820** | **3200** | **6400** |

10.10 Appendix H: Glossary of Terms

Term | Definition |

||| | **RAG** | Retrieval-Augmented Generation; combining retrieved context with LLM generation | | **Grounding** | Traceability of generated suggestions to retrieved source material | | **Reranking** | Re-scoring and reordering retrieved candidates using composite scoring functions | | **Diff** | Code difference between PR branch and base; shows added/removed/modified lines | | **Violation** | An instance of code that doesn't follow specified guidelines; belongs to one of 5 categories | | **True Positive (TP)** | Model correctly detected a violation present in ground truth | | **False Positive (FP)** | Model detected violation not present in ground truth (over-prediction) | | **False Negative (FN)** | Model missed a violation present in ground truth (under-prediction) | | **Line Localization** | Accuracy of pinpointing exact line number where violation occurs | | **Non-Response Rate** | Percentage of inputs where model fails to produce valid output (empty/malformed JSON) | | **Micro Averaging** | Computing metrics by treating all predictions as one pool (opposed to macro averaging) |

10.11 Appendix I: Code Snippets for Common Tasks

1.1 Query the Retrieval Corpus Directly

```
def query_corpus(query_text, top_k=5, category_filter=None):
    """Query retrieval corpus for related guidelines"""
    from src.rag_model.retrieval import SemanticRetriever

    retriever = SemanticRetriever()
    results = retriever.search(
        query=query_text,
        top_k=top_k,
        filters={'category': category_filter} if category_filter else None
    )

    return results

# Example usage
results = query_corpus(
    "naming convention for function parameters",
    top_k=5,
    category_filter="naming_convention"
)
for result in results:
    print(f"Score: {result['score']:.3f}, Text: {result['text'][:100]}...")
```

1.2 Run Static Tool Analysis on a File

```
def analyze_with_static_tools(file_path):
    """Run Pylint and Flake8 on a single file"""
    from src.evaluation.static_tools import StaticToolEvaluator

    evaluator = StaticToolEvaluator(version='v2')
    findings = evaluator.analyze(file_path)

    # findings structure:
    # {
    #   'pylint': [{'line': 10, 'code': 'W0611', 'message': '...', 'category':
    'unused_import'}, ...],
    #   'flake8': [{'line': 5, 'code': 'E501', 'message': '...', 'category':
    None}, ...]
    # }

    return findings

# Example
findings = analyze_with_static_tools('src/module.py')
print(f"PyLint findings: {len(findings['pylint'])}")
print(f"Flake8 findings: {len(findings['flake8'])}")
```

1.3 Generate Review for a Single PR

```
def review_single_pr(pr_url):
    """Analyze a single PR with all three models"""
    from src.rag_model.pipeline import RAGPipeline
    from src.evaluation.static_tools import StaticToolEvaluator
    from src.evaluation.naive_llm import NaiveLLMEvaluator

    # Fetch PR diff
    from src.deployment.github_api import GitHubAPI
    github = GitHubAPI()
    pr_data = github.fetch_pr(pr_url)

    # Run models
    static_results = StaticToolEvaluator(version='v2').evaluate(pr_data)
    naive_results = NaiveLLMEvaluator().evaluate(pr_data)
    rag_results = RAGPipeline().evaluate(pr_data)

    # Aggregate results
    combined = {
        'pr_url': pr_url,
        'static_findings': static_results['findings'],
        'naive_llm_findings': naive_results['findings'],
        'rag_findings': rag_results['findings'],
        'grounding': rag_results.get('grounding', [])
    }
```

```
return combined
```

```
# Example
```

```
results = review_single_pr('https://github.com/owner/repo/pull/123')  
print(f"Static tool found {len(results['static_findings'])} violations")  
print(f"RAG found {len(results['rag_findings'])} violations with {len(results  
['grounding'])} grounded")
```

10.12 References to External Resources

Qdrant Vector Database - Docs: <https://qdrant.tech/documentation/> - GitHub: <https://github.com/qdrant/qdrant>

Sentence Transformers (Embeddings) - Docs: <https://www.sbert.net/> - Model Hub: <https://huggingface.co/sentence-transformers> - Paper: <https://arxiv.org/abs/1908.10084>

FastAPI (Web Framework) - Docs: <https://fastapi.tiangolo.com/> - Tutorial: <https://fastapi.tiangolo.com/tutorial/>

Groq API (LLM Inference) - Console: <https://console.groq.com/> - API Docs: <https://console.groq.com/docs>

GitHub API - REST API: <https://docs.github.com/en/rest> - GraphQL: <https://docs.github.com/en/graphql> - Rate Limits: <https://docs.github.com/en/rest/overview/resources-in-the-rest-api>

Appendix End

All appendices and reference materials have been compiled to support reproducibility, implementation, and future extensions of this project. For questions or clarifications, refer to the specific milestone reports or contact the Group 1 DSAI Lab team.